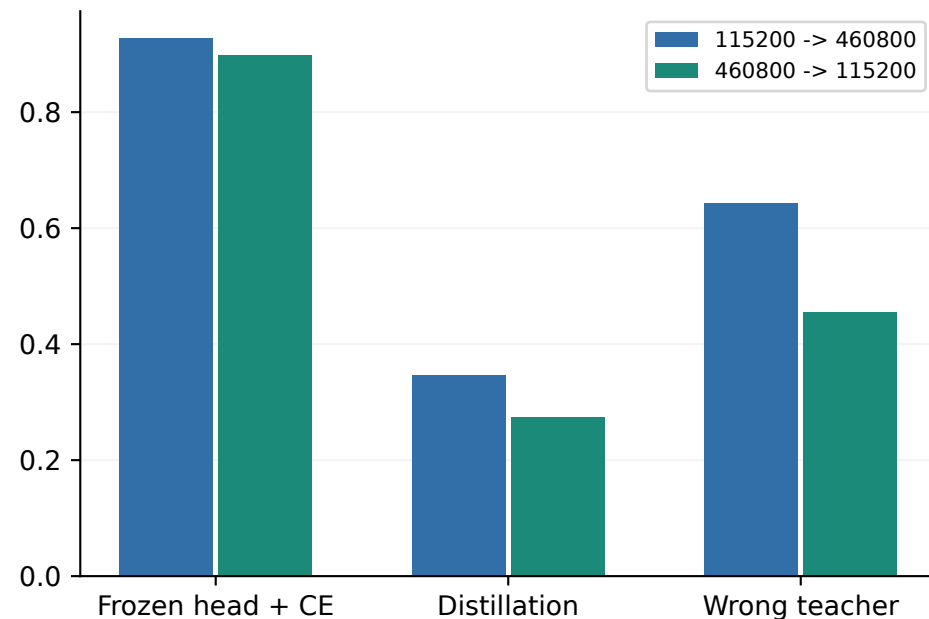
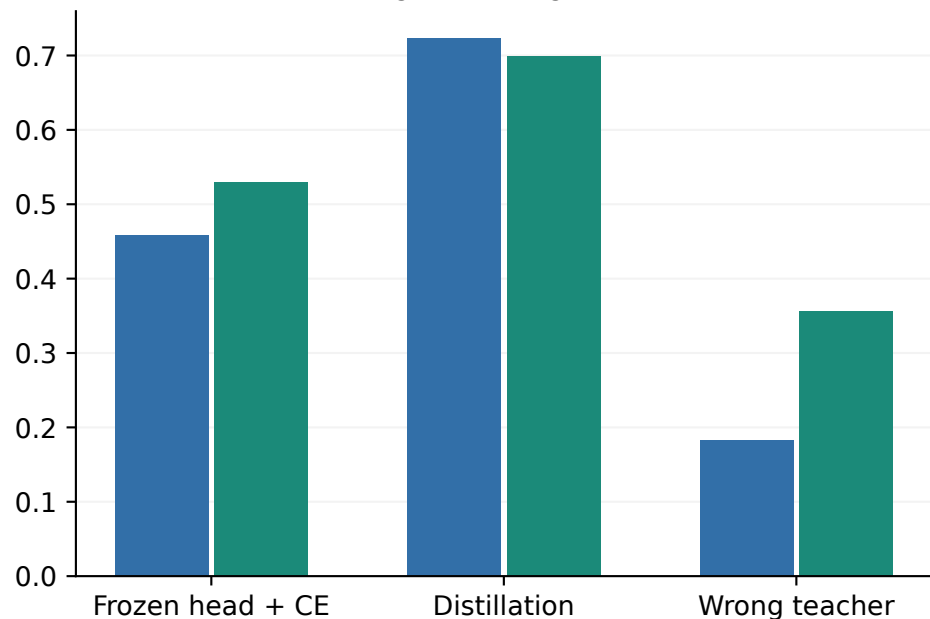


Feature alignment improves; this does not establish precise time alignment

Contact-feature MSE (lower is better)



Paired-bag cosine (higher is better)



Seed 42. One test recording per class per direction: retrieval can reflect class matching alone.